

Comparative Analysis of Machine Learning Algorithms for Personality Prediction

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Abstract

This project focuses on a comparative analysis of machine learning algorithms for personality prediction using two different datasets. The first dataset is the MBTI Personality dataset. It consists of two columns, one named 'type' representing personality labels like INFP, ENFJ etc, and the other named 'posts' containing user generated social media text. It was used for text based personality predictions. While the second dataset is the Young People Survey Dataset. It consists of multiple columns representing behavioral traits, preferences, and lifestyle choices like music, hobbies, habits, etc. All the values in this dataset are numerical or categorical. As it includes structured survey responses reflecting behavioral traits, it was used in structured data-based behaviour prediction. Three classification algorithms, Naive Bayes, Logistic Regression and Support Vector Machine (SVM), were implemented on both the datasets. Preprocessing techniques such as Cleaning and TF-IDF Vectorization were applied for the text data whereas structured dataset required handling missing values and encoding categorical features. The models were assessed using standard performance metrics like accuracy, precision, recall, F1-score and confusion matrix. The results suggest that Logistic Regression performed best on text based dataset while SVM was most accurate on the structured dataset. This report highlights how dataset characteristics influence model performances and shows that no single algorithm is universally optimal. This work signifies the importance of selecting the right models based on the nature of the data for effective personality prediction.

Introduction

Personality prediction is a crucial area of study which focuses on understanding human behaviour, preferences and traits using computational methods. Now with the increased use of social media and digital platforms, a very large amount of user generated data is now available, which can be analysed to understand and gain insights into individual personalities. Machine learning techniques help process such data and identify patterns that may not be easily observable by traditional methods.

Different machine learning algorithms work differently depending on the type of the data being used. For instance, text based data, such as social media posts, many times contains noise and requires preprocessing techniques whereas structured data, such as survey responses, is typically more organised and easier to analyze. Hence, it becomes important to compare the performance of all multiple algorithms across different types of datasets.

In this project, three algorithms, Naive Bayes, Logistic Regression, Support Vector Machine (SVM), are applied to two different datasets, a text based MBTI dataset and a structured survey dataset. The main purpose of this project is to evaluate and compare their performance using various metrics such as accuracy, precision, recall and F1 score.

This study aims to highlight how the nature of the data used influences model performance and identify the most suitable algorithm of personality prediction tasks.

Objectives

- To implement machine learning algorithms for personality prediction
- To apply Naive Bayes, Logistic Regression, and SVM on different datasets
- To compare the performance of these models using evaluation metrics
- To analyze the impact of data type (text vs structured) on model performance
- To identify the most suitable algorithm for personality prediction

Methodology

Datasets

In this project, two different types of datasets were used in analysing the performance of the machine learning algorithms on personality predictions tasks.

The first dataset is the MBTI (Myers-Briggs Type Indicator) Personality dataset, that contains user-generated social media posts with their equivalent or matching personality types based on the Myers-Briggs type indicator. It consists of two main columns, one representing the personality type, like INTP, ENFP, etc, and the other containing textual data in the form of posts. This dataset is mostly unstructured and text based in nature.

The second dataset is the Young People Survey Dataset, that contains structured survey responses collected from individuals. It includes various features connected to lifestyle, habits, behaviours and preferences. For this project though, the “Alcohol” column was selected as the target variable, representing the behavioral traits. The dataset contains both numerical and categorical values which makes it suitable for structured data analysis.

Data Preprocessing

Since the two datasets are of different types, different preprocessing techniques were applied:

For MBTI Dataset which is text data:

- All the text was converted to lowercase
- URLs and special characters were removed
- Extra spaces were cleaned
- The cleaned text was transformed into numerical features using TF-IDF Vectorization
- A maximum of 5000 features was used to balance performance and computation.

For Survey Dataset which is structured data:

- Only numerical features were selected for model training
- Missing values were handled by replacing them with the median of each column
- The target column, Alcohol, was encoded into numerical values using label encoding
- Feature scaling was applied using StandardScaler for Logistics Regression and SVM

Algorithms Used

Three machine learning algorithms were implemented and compared:

Naive Bayes: A probabilistic classifier based on Bayes theorem.

- Multinomial Naive Bayes was used for text data
- Gaussian Naive Bayes was used for structured data

Logistic Regression: A linear classification algorithm used for predicting categorical outcomes. It performs well on both text and structured data when features are properly scaled.

Support Vector Machine (SVM): A powerful classification algorithm that finds the optimal hyperplane to separate classes. LinearSVC was used for faster computation.

Model Training and Testing

- The datasets were divided into training and testing sets using 80:20 split.
- Stratified sampling was used to maintain class balance
- Models were trained on the training data
- Predictions were made on the testing data

Evaluation Metrics

To evaluate the performance of the models, below metrics were used:

- **Accuracy** – Measures overall correctness of the model
- **Precision** – Measures how many predicted positives are actually correct
- **Recall** – Measures how many actual positives were correctly identified
- **F1-score** – Harmonic mean of precision and recall
- **Confusion Matrix** – Shows correct and incorrect predictions for each class

Visualization

To better understand model performance, visualizations were created: Bar graphs to compare accuracy across models, Combined comparison chart for both datasets and Confusion matrix heatmap for the best performing model.

Results

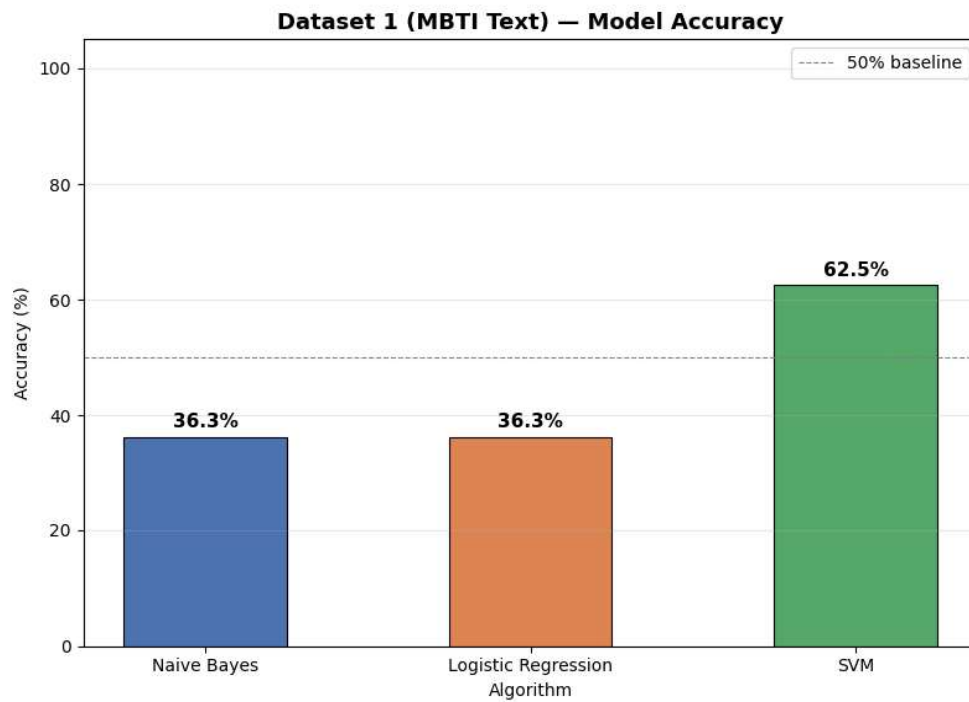
This section contains the performance of the implemented machine learning models on both the datasets.

Results on MBTI Dataset

Logistic Regression achieved the highest accuracy of 62.54%, making it the best performing model for the text based personality prediction. SVM stands on a close second as it performed well with similar results. Naive Bayes showed significantly lower performance due to its simplifying assumptions.

The overall accuracy is moderate as the dataset contains 16 personality classes which makes the classification complex.

Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	36.31	42.63	36.31	28.10
Logistic Regression	62.54	63.88	62.54	59.82
Support Vector Machine (SVM)	60.17	60.08	60.17	59.38

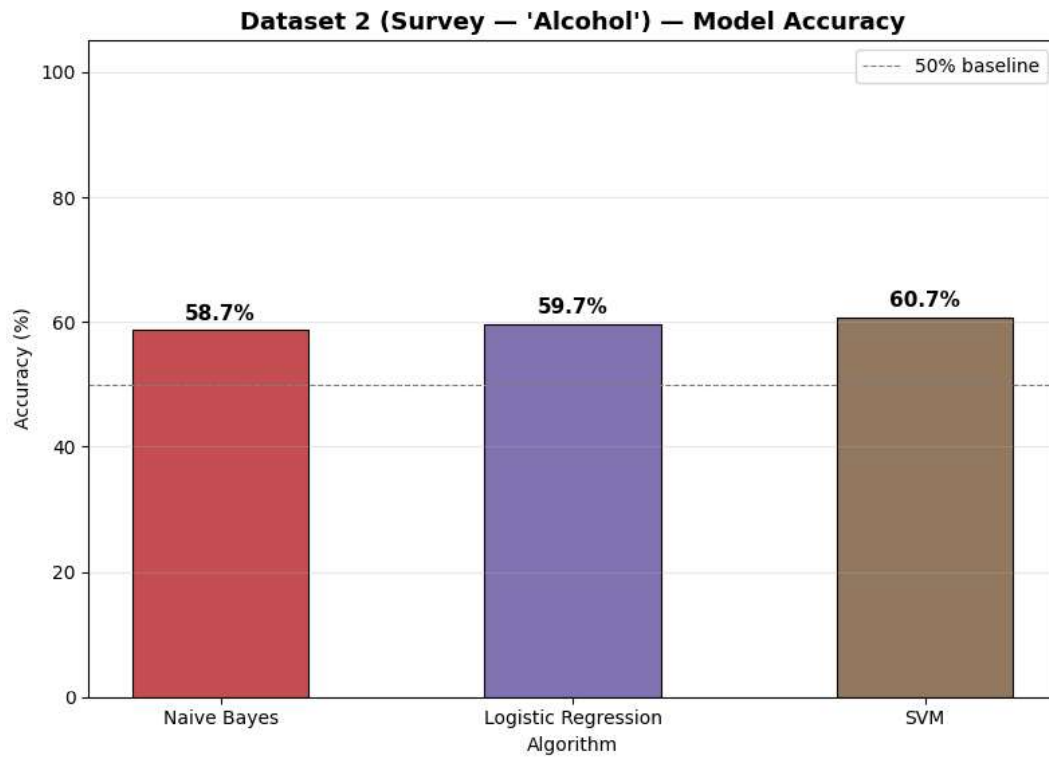


Results on Survey Dataset

SVM achieved the highest accuracy of 60.70%, which makes it the best performing among the three for structured data. Logistic Regression closely follows behind it with similar results. Naive Bayes performed reasonably well due to the numerical nature of the data.

The overall performance is better compared to the text dataset due to cleaner and more structured features.

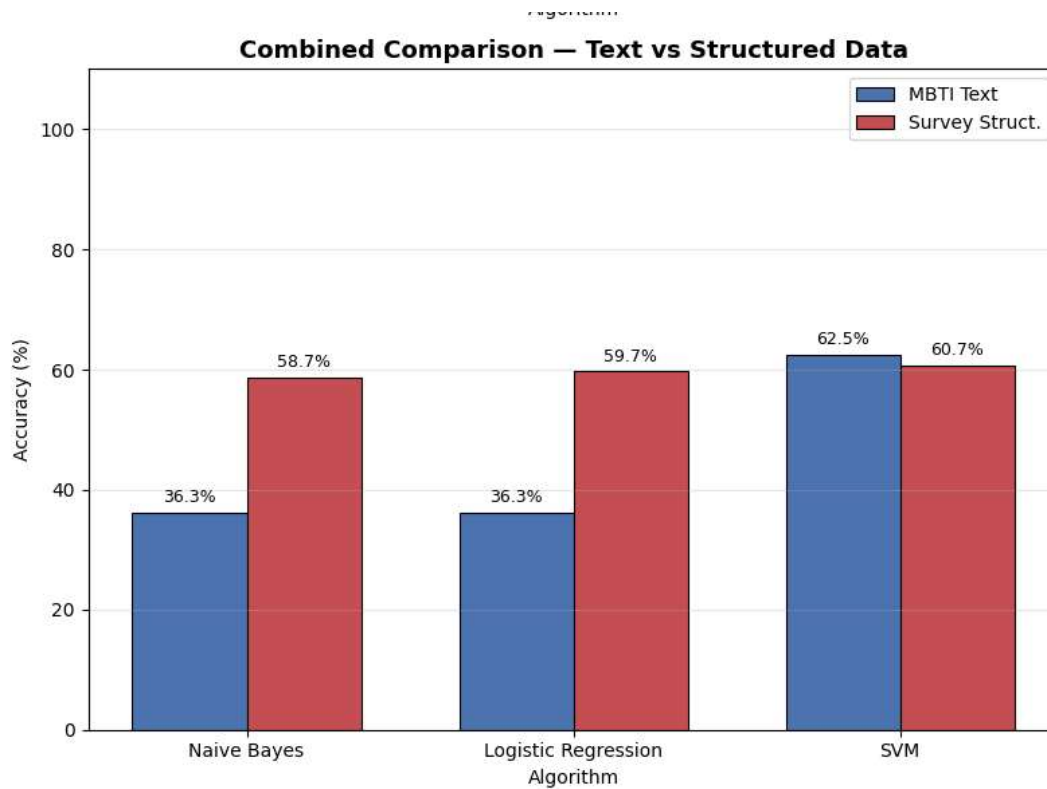
Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	58.71	59.56	58.71	59.09
Logistic Regression	59.70	60.34	59.70	60.00
Support Vector Machine (SVM)	60.70	58.25	60.70	59.35



Combined Comparison

SVM achieved the highest accuracy of 60.70%, which makes it the best performing among the three for structured data. Logistic Regression closely follows behind it with similar results. Naive Bayes performed reasonably well due to the numerical nature of the data.

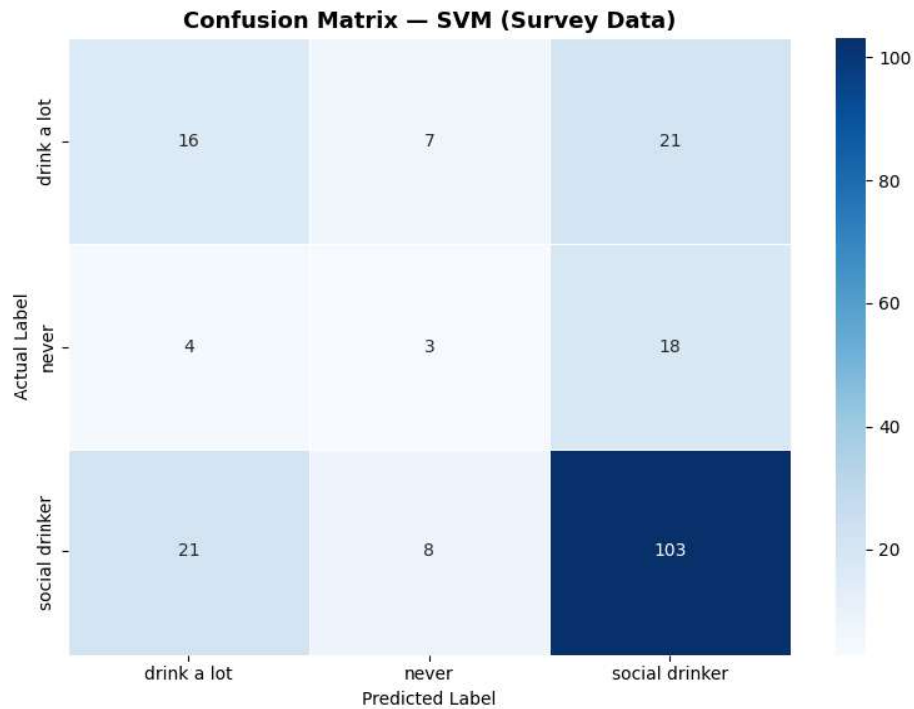
The overall performance of the Survey dataset is better compared to the text dataset due to cleaner and more structured features.



Confusion Matrix

A confusion matrix was generated for the best-performing model on the survey dataset (SVM) to visualize prediction performance.

- The confusion matrix shows how many predictions were correct and incorrect
- Diagonal values = correct predictions
- Off-diagonal values = misclassifications



Discussion

The results obtained from both the datasets give important insights into the performance of different machine learning algorithms on personality prediction tasks.

Among the three models, Logistic Regression performed the best on the MBTI text dataset which was unstructured, achieving the highest accuracy. That is because Logistic Regression handles high-dimensional sparse data, such as TF-IDF features, more effectively compared to Naive Bayes. Additionally, it does not assume feature independence, which allowed it to capture more complex relationships in the dataset.

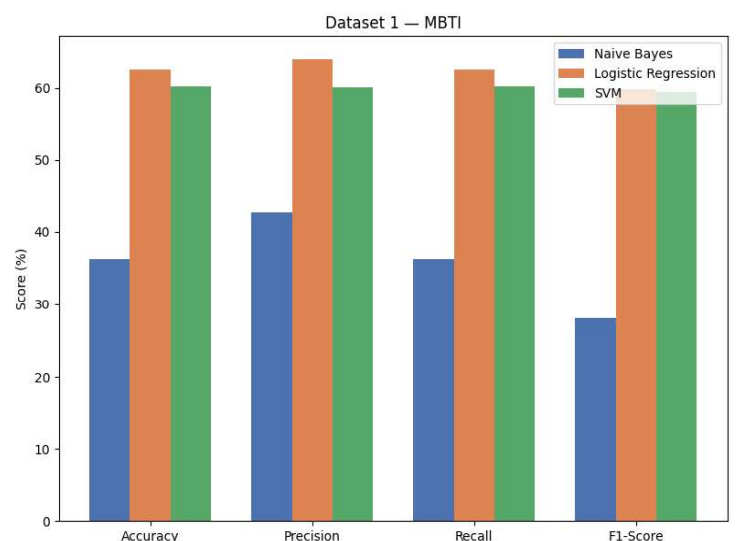
On the other hand, SVM performed best on the structured survey dataset. That can be linked to the fact that SVM works well with clearly defined numerical features and is effective in finding optimal decision boundaries in structured data. The use of feature scaling improved its performance more.

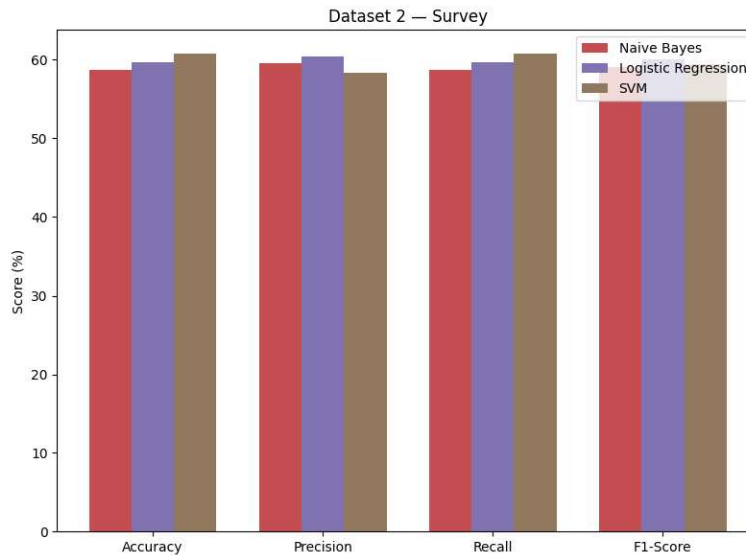
The worst performing model Naive Bayes on the MBTI dataset. This is mostly due to it's strong assumption that all the features are independent of each other, which is not true for natural language data where even words have contextual relationships. Therefore, its accuracy was significantly lower compared to the other models.

Another significant observation is that models performed better on structured data than on the unstructured text data. Structured datasets are cleaner and contain well-defined numerical values, making it easier for models to learn patterns. On the contrary, text data is noisy, unstructured, and high-dimensional, which increases the complexity of the prediction task.

It is also important to note that high accuracy does not always indicate a reliable model. In classification tasks, especially with multiple classes or imbalanced datasets, accuracy alone can be misleading. Metrics such as precision, recall, and F1-score provide a more comprehensive evaluation of model performance.

Altogether, the results highlight that the effectiveness of a machine learning algorithm depends heavily on the nature of the dataset. No single model performs best in all scenarios, and selecting the appropriate algorithm requires understanding the characteristics of the data.





Conclusion

This project offered a comparative analysis of three machine learning algorithms for personality prediction using both text-based and structured datasets. The results showed that Logistic Regression performed best for text data, while SVM achieved the highest accuracy for structured data.

The study shows that the performance of machine learning models is highly dependent on the type and quality of the dataset. Structured data generally leads to better performance due to its clarity and organization, whereas unstructured data introduces complexity due to noise and high dimensionality.

Limitations

Limitations of the current project is that no deep learning models were used. Datasets used were moderate sized, advanced models and systems could be used on much larger datasets in order to get better results.

Future Improvements

- Use advanced NLP techniques
- Apply deep learning models (like LSTM, BERT)
- Improve feature selection and preprocessing

Implementation/Code Access

The complete implementation of this project, including data preprocessing, model training, evaluation, and visualization, has been carried out using Google Colab. The full code can be accessed using the link below:

 IIS_Mini_Project_3(final).ipynb

References

Kaggle – MBTI Personality Dataset

<https://www.kaggle.com/datasnaek/mbti-type>

Kaggle – Young People Survey Dataset

<https://www.kaggle.com/miroslavsabo/young-people-survey>

Scikit-learn Documentation

<https://scikit-learn.org/stable/>